

Bangladesh University of Engineering and Technology

Department of Chemistry

CHEM113: Chemistry

For Students of CSE

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CHEM113 : Chemistry

Total Credit Hours: 3 ; L1/T2

Course Content

Quantum concept in atomic structure, VSEPR; molecular geometry, Quantum concept in bonding; VBT and MOT, Frontier MOT and electronic transition, Silicon chemistry, Properties of solutions, Colloid and Nano-chemistry, ~~Phase rule and phase diagram; Energy and chemistry, Electrochemistry; electrolytic conduction, corrosion, devices for energy storage, Chemistry of biodegradable and conductive polymer;~~ LED, LCD/touch screen, Chemistry of proteins, nucleic acids (DNA, RNA), carbohydrates and lipids; Introduction to computational chemistry; Design of new molecules, materials and drug.

LED, LCD/touch screen

LED : Liquid Crystal Display

Importance, Liquid crystal (LC), Origin of liquid crystallinity,
Principle of LCD

LED: Light Emitting Diode

Importance, Fundamental of LED, Principle of touch screen
display

LED vs LCD

Liquid Crystal Display: LCD

1971

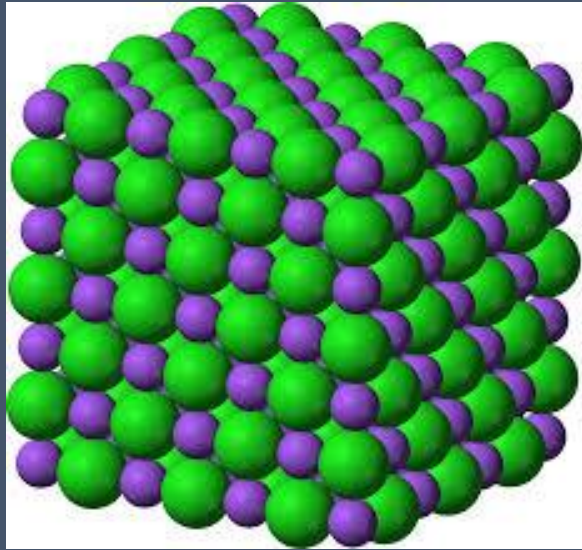
Martin Schadt and
Wolfgang Helfrich



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Liquid Crystal Display: LCD



A crystal is a solid material where atoms are arranged in a highly ordered, repeating pattern, forming a crystal lattice. This ordered structure gives crystals their distinct shapes and properties.

Ordered Structure: Crystals are characterized by a regular, repeating arrangement of atoms, ions, or molecules in a three-dimensional structure. This arrangement is called a crystal lattice.

Anisotropic behavior

Optical anisotropy, in simple terms, means that a material's optical properties, like how it transmits or reflects light, are different depending on the direction of the light passing through it. This is often observed in crystals and other materials with a non-uniform structure

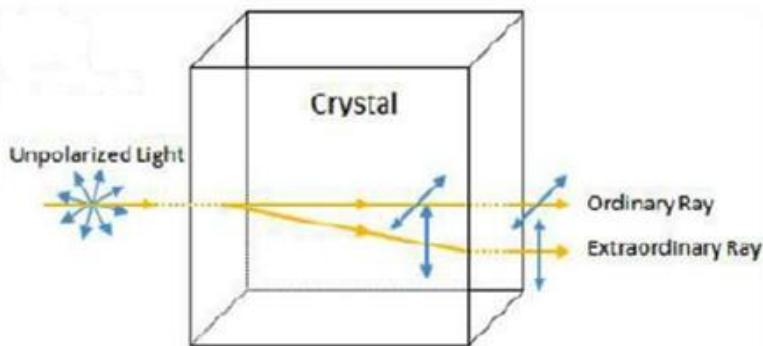
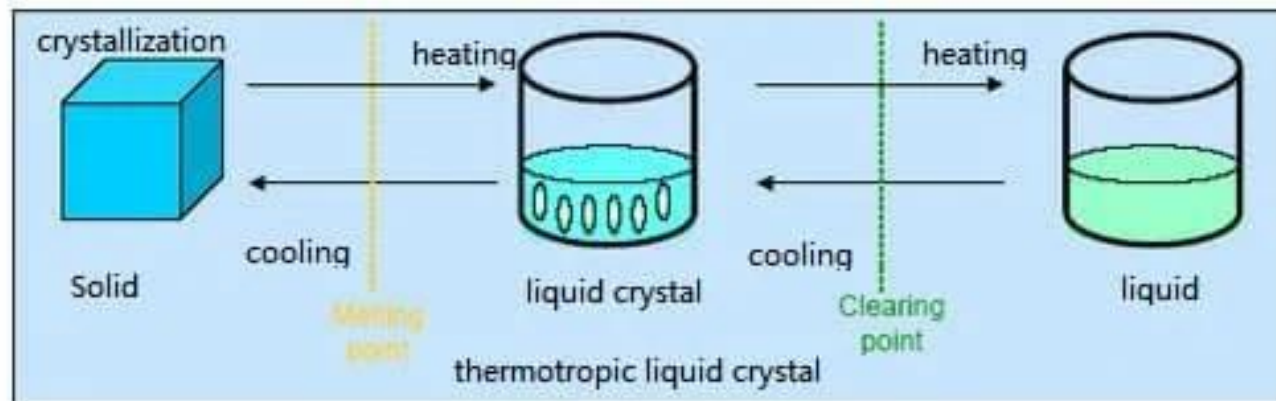
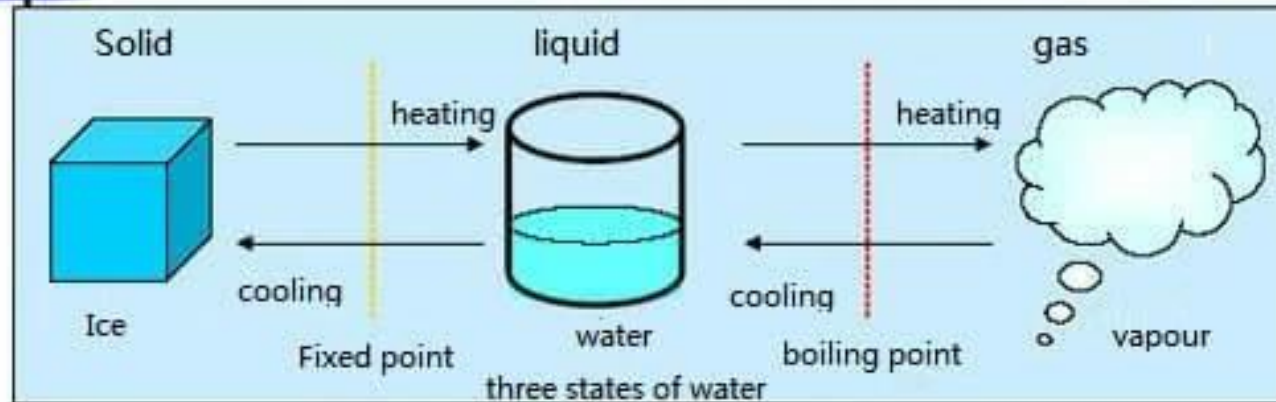


Figure 2. Light passing through a crystal.

Liquid but Crystalline: LCs

Phase change of state



Isotropic : General liquid

Anisotropic : Liquid crystal

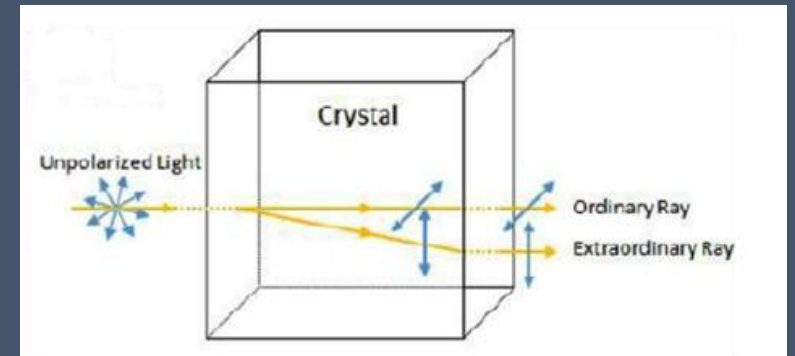
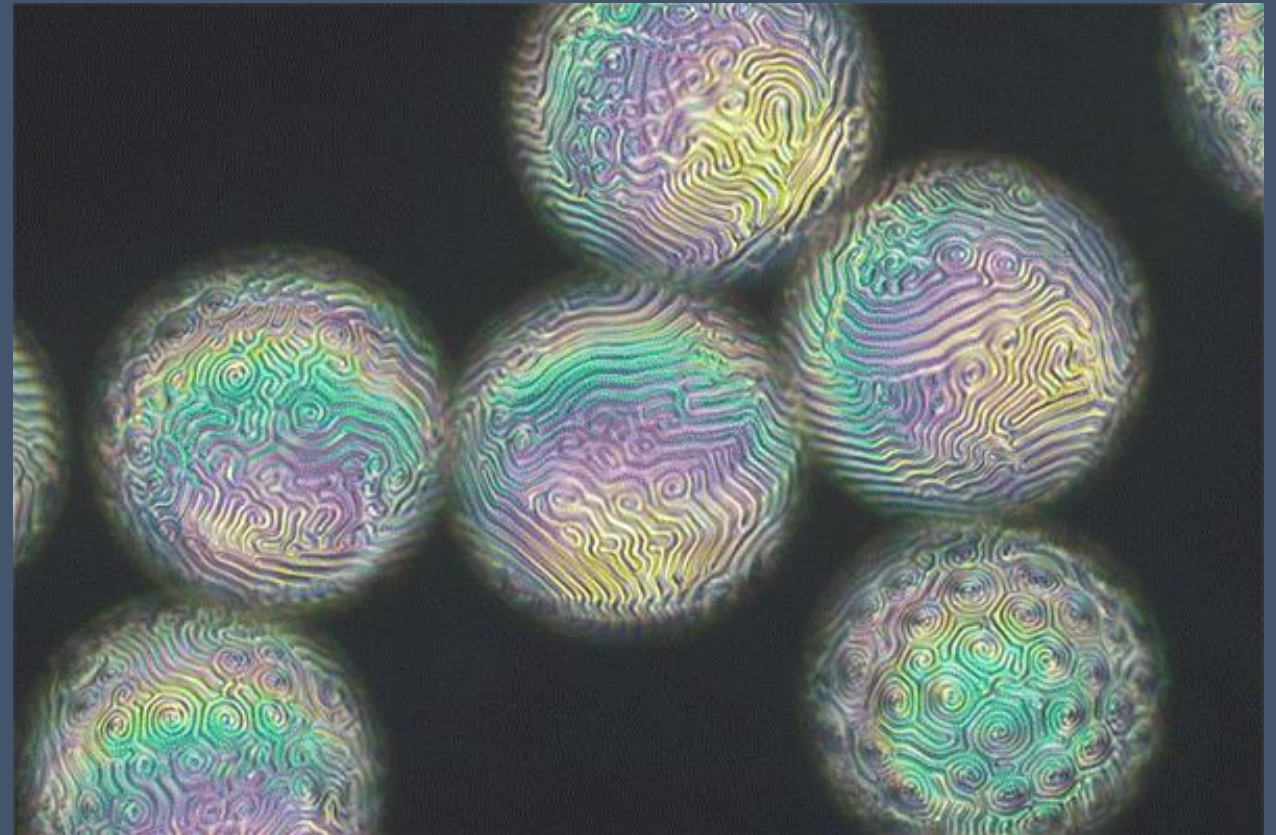
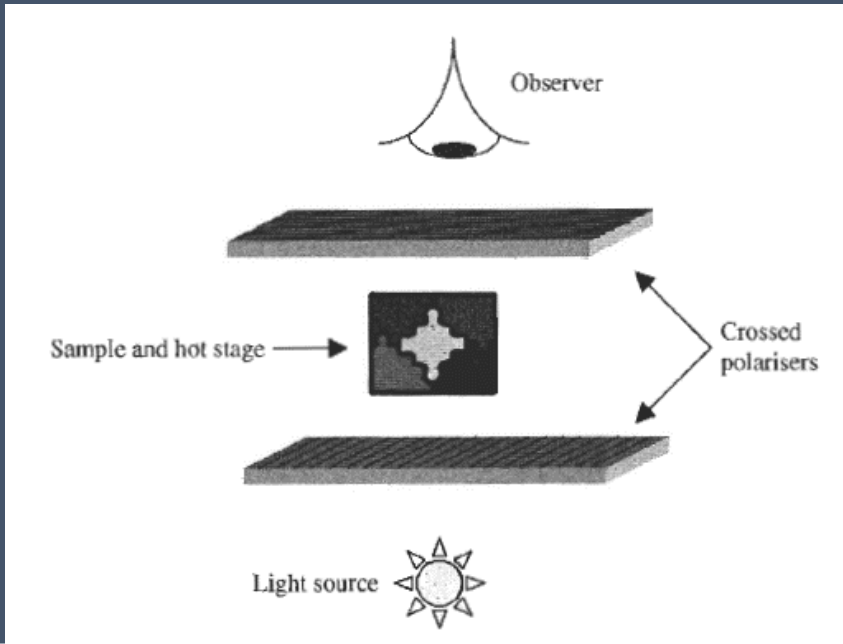
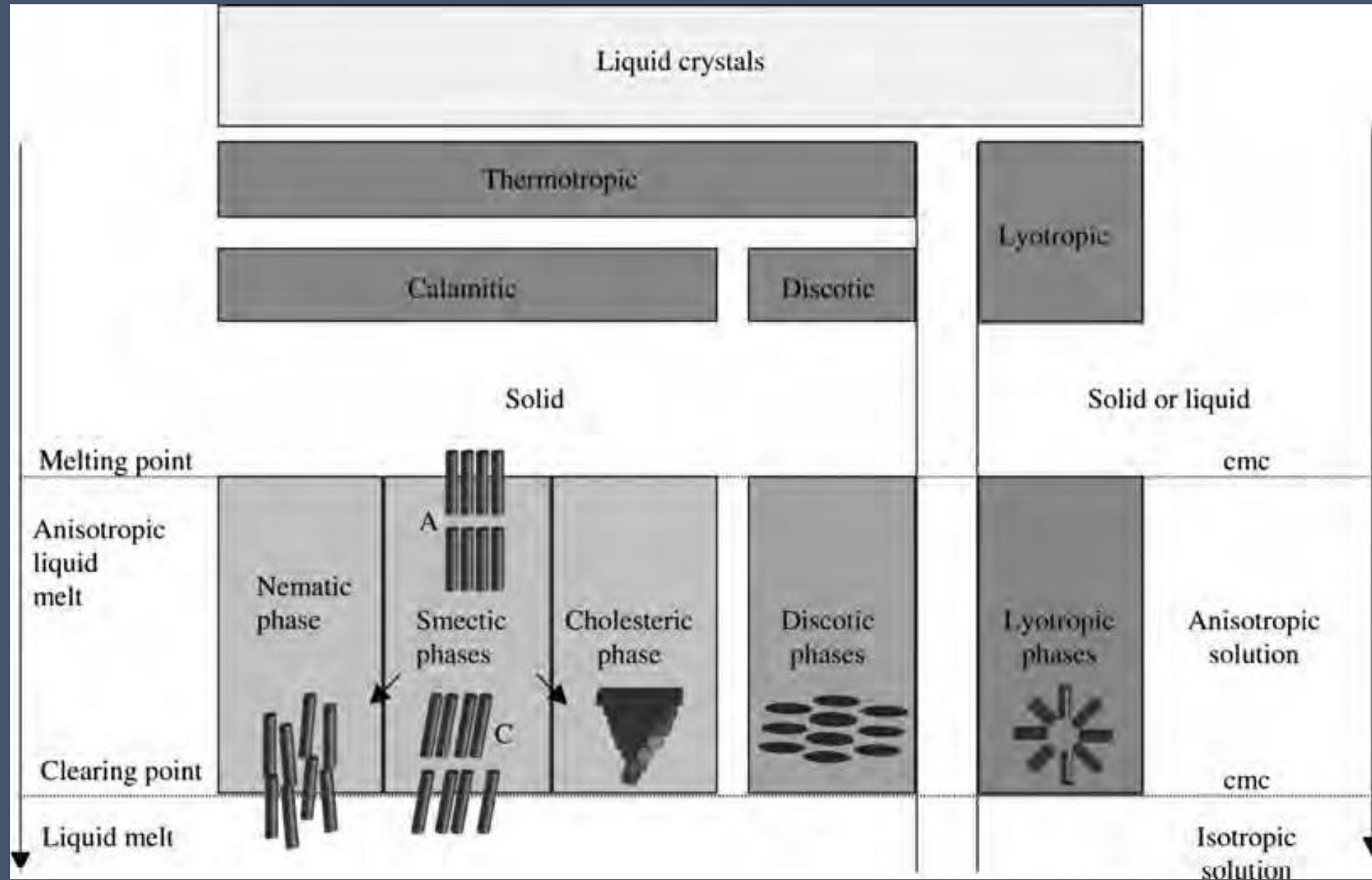


Figure 2. Light passing through a crystal.



Operation of a polarising hot-stage microscope.

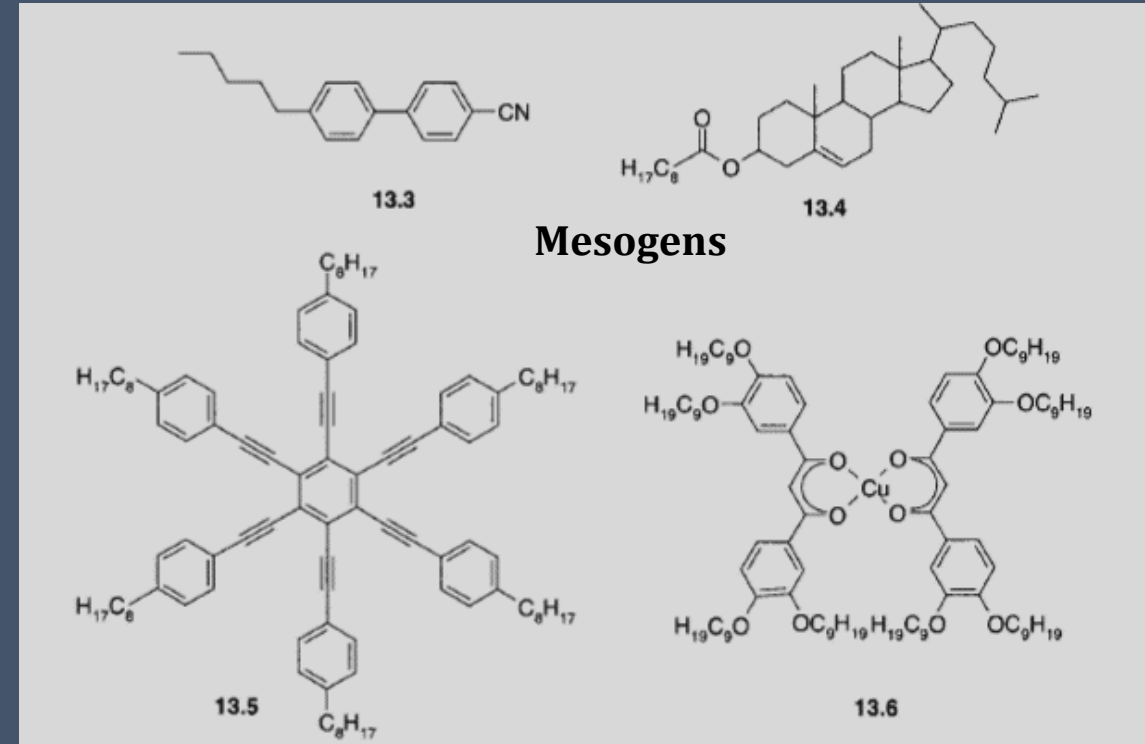
Mesogens and Classification of LCs



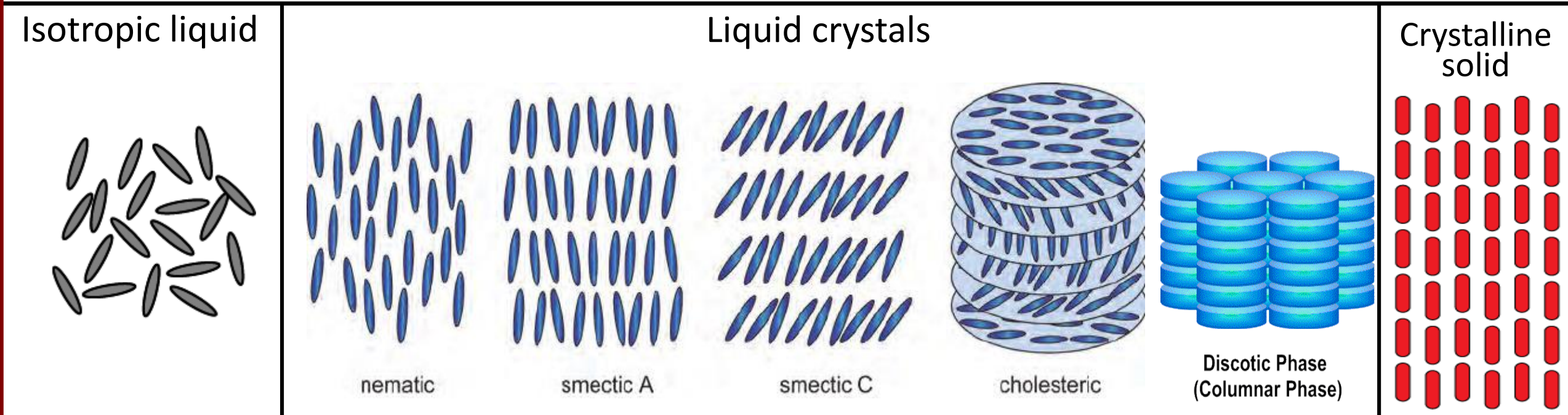
Liquid crystalline materials: Features of Mesogens

A mesogen is a molecule that can exhibit liquid crystalline properties, meaning it can form a mesophase, a state of matter with properties between those of a crystalline solid and an isotropic liquid.

- ❑ The molecular shape should be relatively thin or flat, especially within rigid molecular frameworks.
- ❑ The molecular length should be at least 1.3 nm (13 Å), consistent with the presence of long alkyl groups on many room-temperature liquid crystals.
- ❑ The structure should not be branched or angular.
- ❑ A highly anisotropic polarisability is required in order to enhance intermolecular dispersive interactions. This is favoured by the presence of easily polarisable groups and permanent dipoles.
- ❑ An extended, structurally rigid, highly anisotropic



Molecular ordering and phase transition



Further reading ...

- One phase differs from another with respect to its symmetry.
- The most fundamental characteristic of a LC is the presence of long-range orientational order while the positional order is either limited (smectic phases) or absent altogether (nematic phases).
- The transition between different phases corresponds to the breaking of some symmetry and can be described in terms of the so-called order parameter.
- The free energy functional is written in terms of the order parameter, which describes the orientation of the LC molecules. For isotropic liquid, the order parameter is zero.

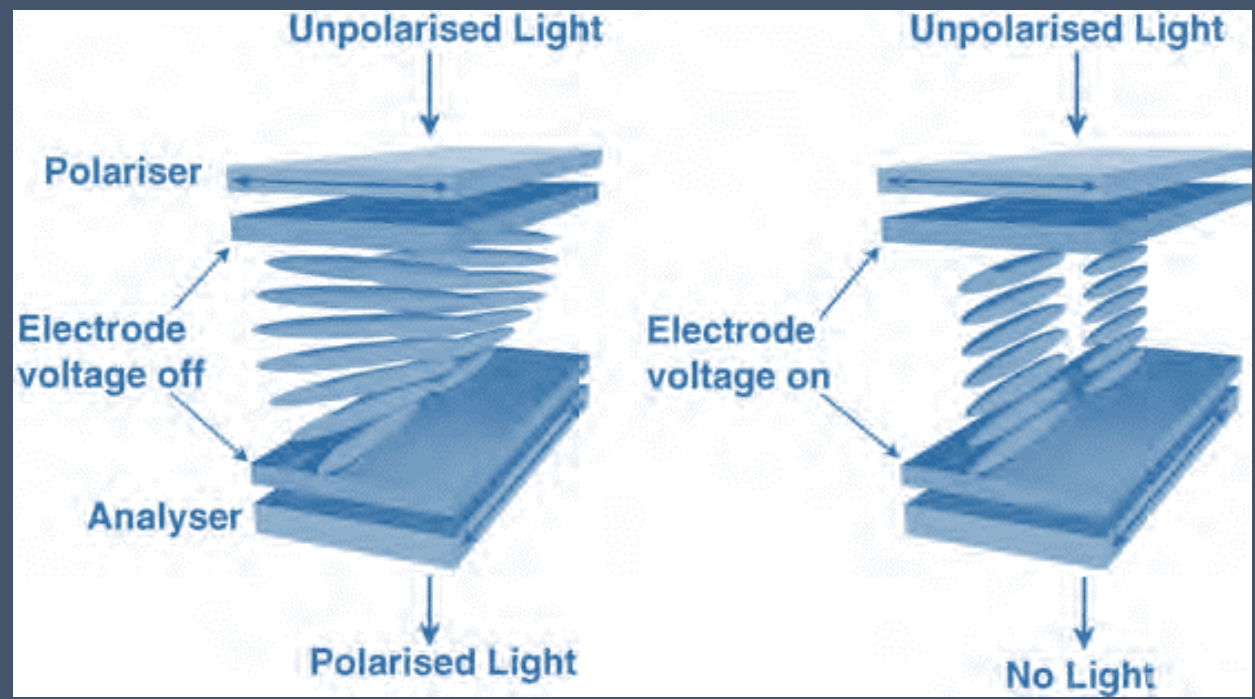
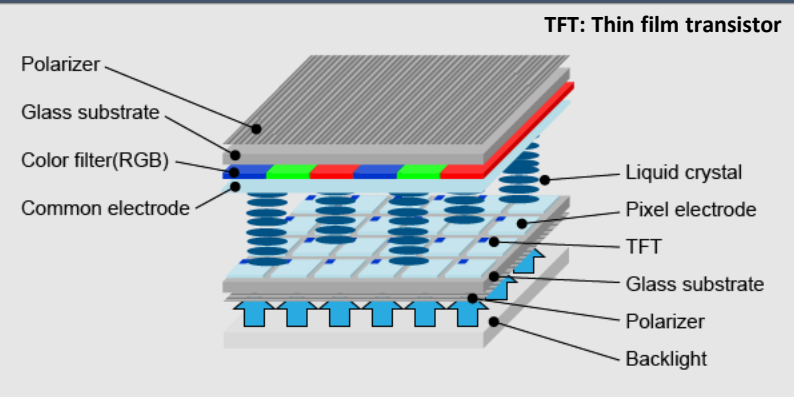
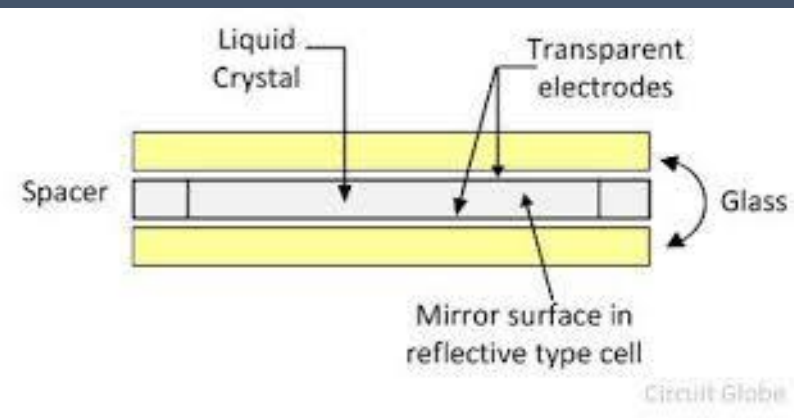
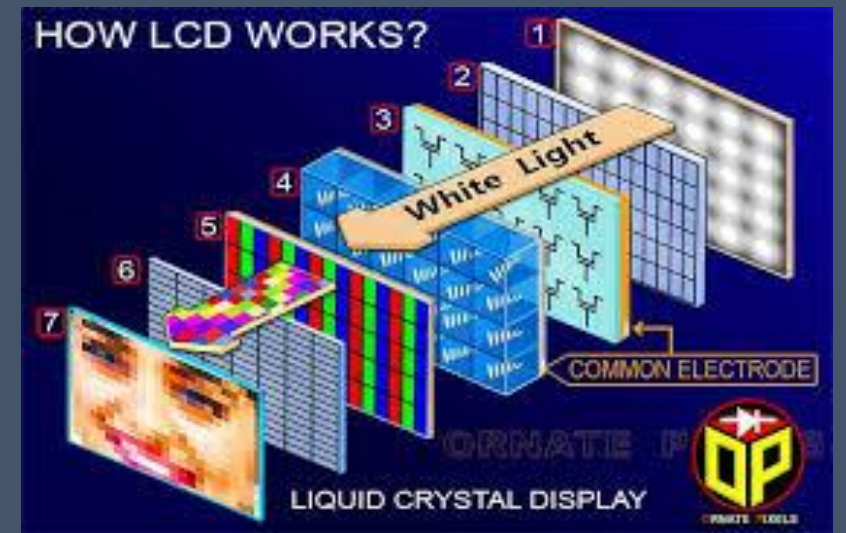
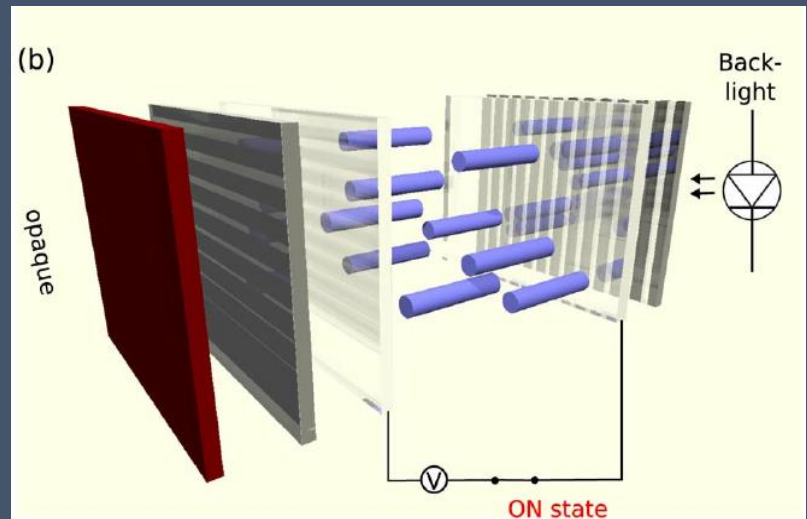
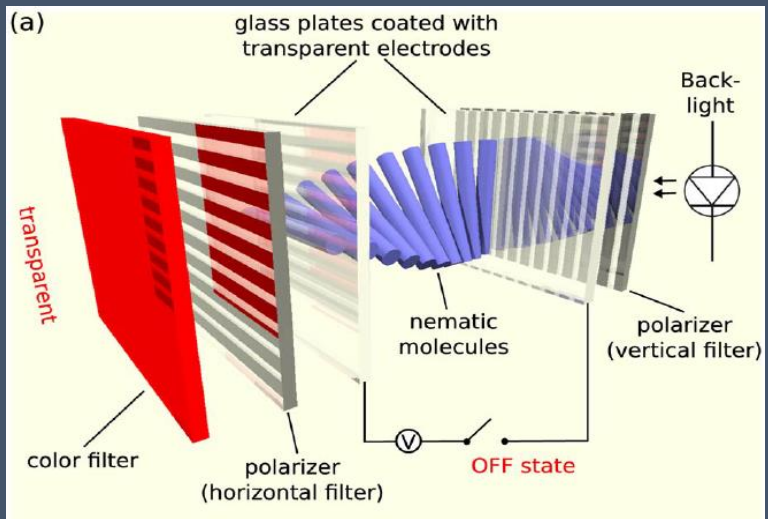


Fig. Operation of a twisted nematic cell. In the absence of an applied electric field, the twisted nematic rotates the plane of polarised light from one side of the cell to the other allowing light to pass through. An applied potential renders it opaque.



LCDs work on the principle that an applied electric field will alter the molecular orientations and result in changes in the optical behaviour. In a TNC, a thin layer (10–15 μm) of nematic LC is sandwiched between two transparent glass plates. The glass plates are pretreated with a director that has an orientating effect on the LC phase, much like the mesogen orientates itself by interactions with its neighbours. If the two plates are rotated at 90 degrees to each another, a helical twist in the nematic phase is produced, with twist angle of 90° from one side to the other (Figure). The polarisation of plane-polarised light passing through the layer follows this 90-degree twist. The orientated liquid crystalline layer is placed between crossed polarisers, resulting in a display that is transparent to light at rest in the absence of an electric field.

Application of a voltage produces a homeotropic rearrangement of the nematic phase and the twist is lost, resulting in no rotation of the plane-polarised light, no light transmission, and hence a darkened display. Because a voltage is required only to darken portions of the display when in use, the power consumption is extremely small, making the TNC ideal for use in portable applications such as watches, portable computers etc. The displays are also flat and compact, and their readability increases with ambient luminosity.

Disadvantages are that the displays cannot be read in the dark (unless backlit), nor at an extreme viewing angle and their response time is slow (about 50 μs) compared with cathode ray tubes (10 μs), limiting their application in television sets, for example. The latest generation of FL at panel displays get round some of these problems by using a patterned vertical alignment (PVA) method. In a conventional PVA a vertically aligned liquid crystal (LC) director tilts down in four directions making 45° with respect to crossed polarisers to give a wide viewing angle. The parallel geometry reduces the lag time while molecules reorientate when the voltage is turned off because the twisting occurs when the voltage is switched on.

More modern, much faster responding liquid crystalline displays use ferroelectric liquid crystals such as p-decyloxybenzylidene-p'-amine-2-methylbutylcinnamate, which can be switched between two stable states by application of an external field. Such properties may also be useful in optical data storage.

Further reading ...

Other Applications of LC

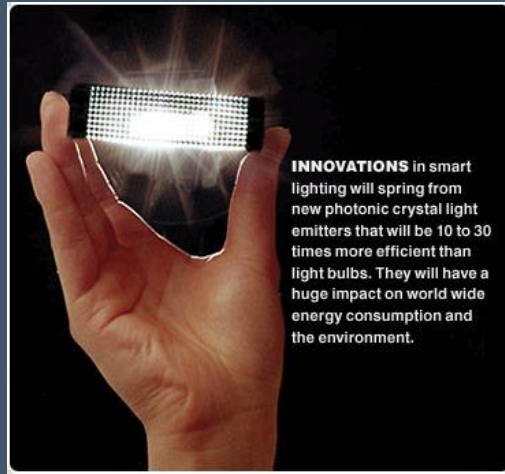
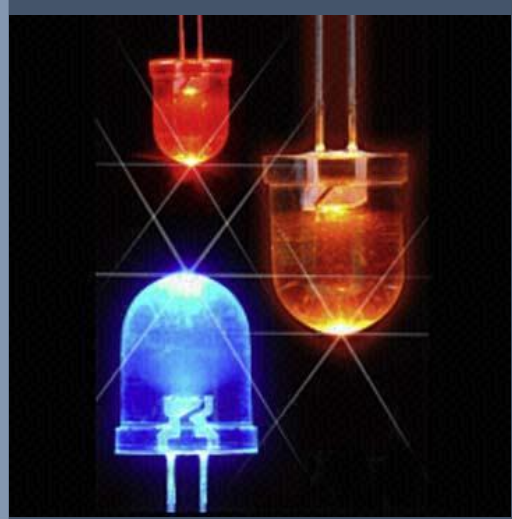
LC have a number of technologically important applications.

- LC thermometers: LC are used in precise temperature measurement, giving a temperature resolution of up to 0.007 K in a 1 K range in the range of $-20\text{ }^{\circ}\text{C}$ to $250\text{ }^{\circ}\text{C}$.**
- LCs are used in various areas of analytical chemistry via measurement of anisotropic properties, such as circular dichroism phenomena, by IR and UV-visible spectroscopy.**
- LCs are used as stationary phases in the chromatographic separation.**
- A range of NMR applications also exist.**

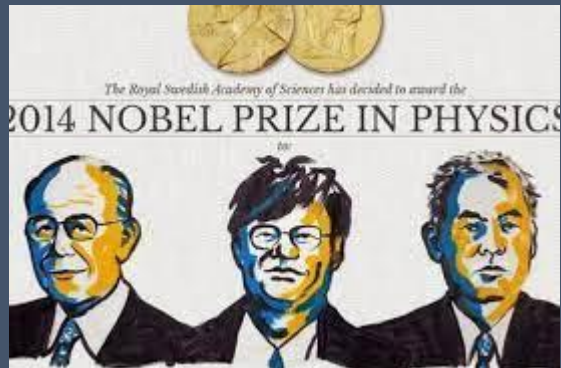


LIGHT EMITTING DIODE (LED)

LED are semiconductor p - n junctions that under forward bias conditions can emit radiation by electroluminescence in the UV, visible or infrared regions of the electromagnetic spectrum. The qaunta of light energy released is approximately proportional to the band gap of the semiconductor.



INNOVATIONS in smart lighting will spring from new photonic crystal light emitters that will be 10 to 30 times more efficient than light bulbs. They will have a huge impact on world wide energy consumption and the environment.



The 2014 Nobel Prize in Physics was awarded to Isamu Akasaki, Hiroshi Amano, and Shuji Nakamura for the invention of efficient blue light-emitting diodes (LEDs)

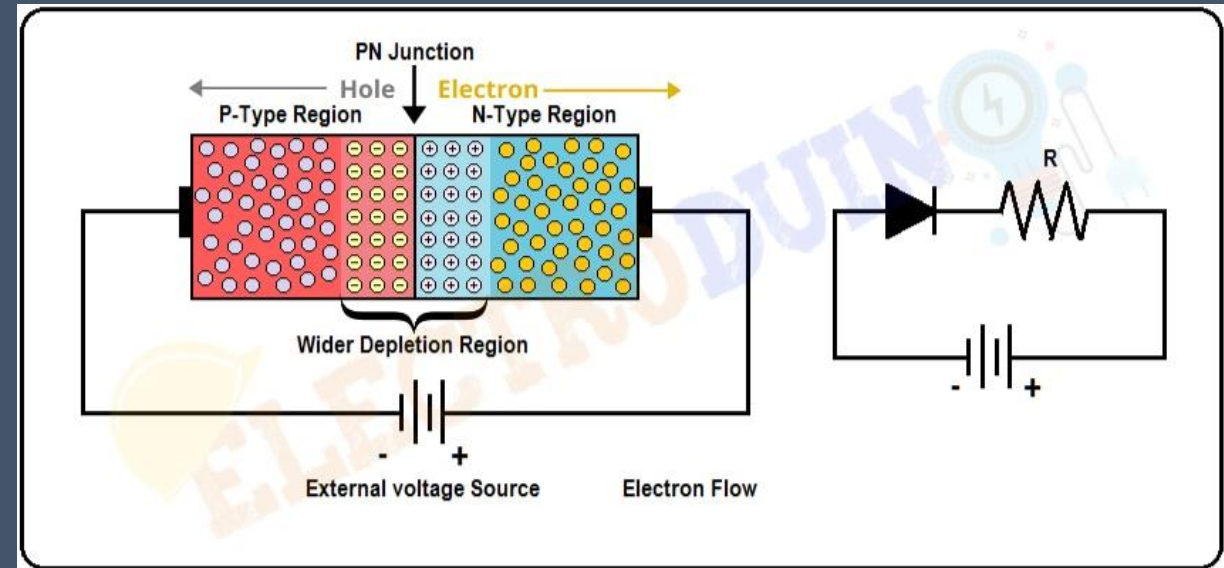


Fundamental of LED

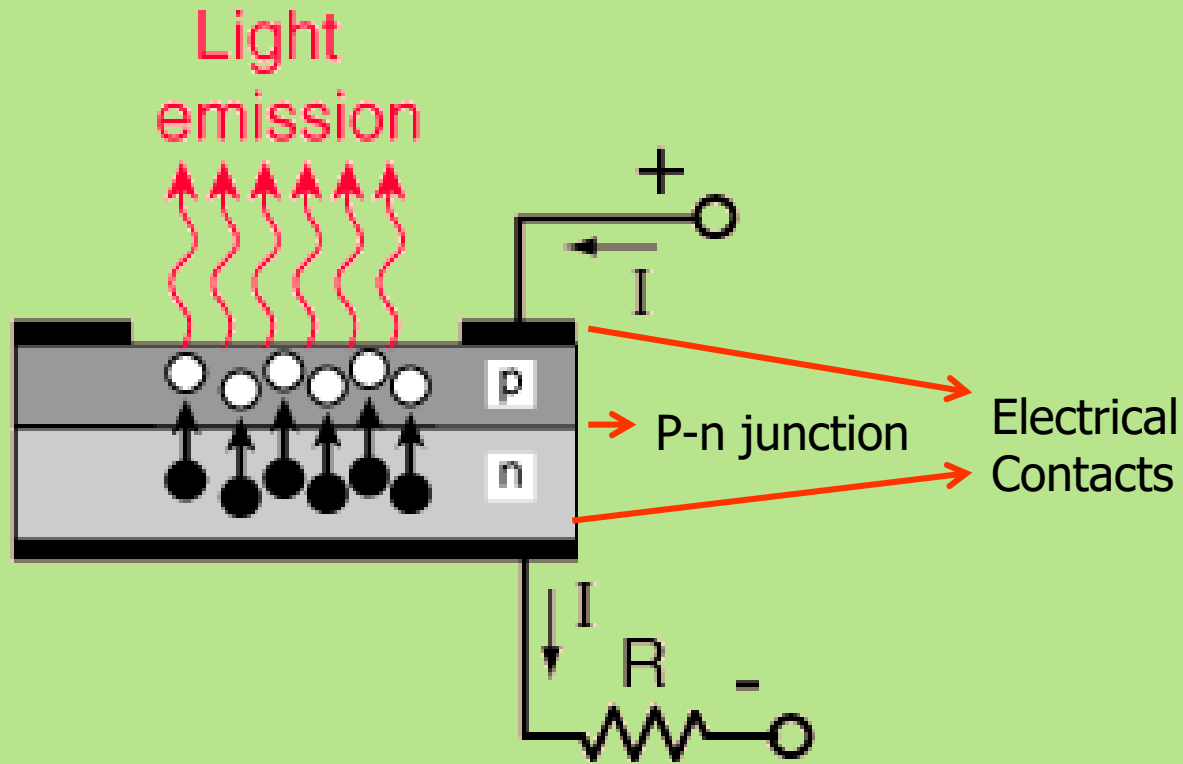
A *p-n* junction diode is a semiconductor device formed by joining a *p*-type and an *n*-type semiconductor material.

It allows current to flow easily in one direction (forward bias) but restricts it in the opposite direction (reverse bias), acting as a one-way valve for electrical current.

This behavior is due to the formation of a depletion region at the junction, which creates a potential barrier that needs to be overcome for current to flow.



How does it work?



A typical LED needs a *p-n* junction

There are a lot of electrons and holes at the junction due to excitations

Electrons from *n* need to be injected to *p* to promote recombination

Junction is biased to produce even more *e-h* and to inject electrons from *n* to *p* for recombination to happen

Recombination produces light!!

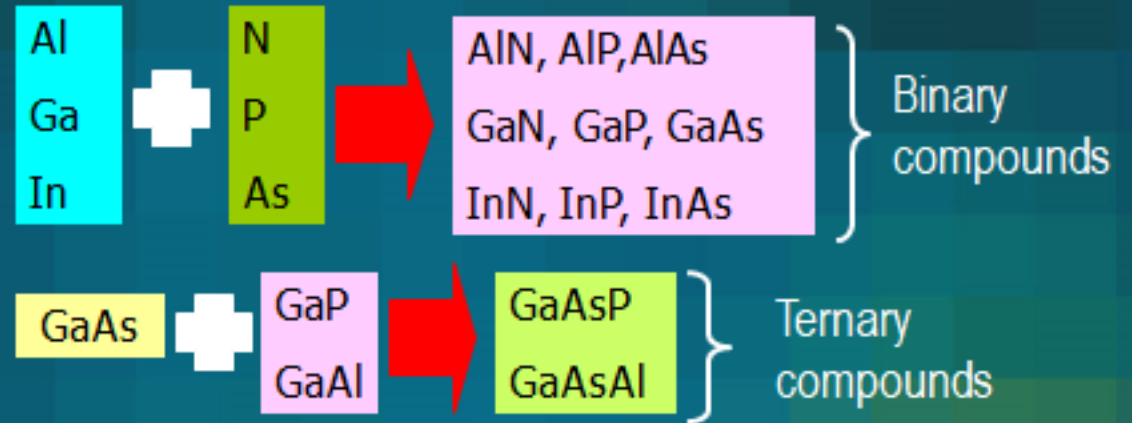


Candidate Materials Group III-V & Group II-VI

Group II		Group III		Group IV		Group V											
				iii	iv	v											
H				B	C	N	O	He									
Li	Be			Al	Si	P	S	Ne									
Na	Mg			Ga	Ge	As	Se	Ar									
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn						
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd						
Cs	Ba		Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	In	Sn	Sb	Te	I	Xe
Fr	Ra		Rf	Db	Sg	Bh	Hs	Mt	Uun	Uuu	Uub						
		La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	
		Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr	

Periodic Table to show group III-V and II-V binaries

Group III-V LED materials



Questions to ask when choosing the right material:

1. Can it be doped or not?
2. What wavelength it can emit?
3. Would the material be able to allow radiative recombination?
4. Direct or indirect semiconductor?

LCD vs. LED



Angle view???

Difference between LED and LCD

LED		LCD
• LED has a better response time than LCD.		☐ LCD is slower than LED in terms of response time.
• LED consumes more power in comparison to LCD.		☐ LCD consumes less power in comparison to LED.
• LED delivers good picture quality in comparison to the LCD display.		☐ LCD also delivers good picture quality but less than LED.
• LED is costlier than LCD.		☐ LCD is less costly than LED.
• LED has a better black level and contrast in comparison to LCD.		☐ LCD has not good black level and contrast as LED.
• LED delivers better color accuracy.		☐ LCD delivers good color accuracy
• LED has a wider viewing angle than the LCD.		☐ In LCD, the wide-angle decreases with 30 degrees from the center in the image then the contrast ratio.
• LEDs use no mercury and are therefore environmentally friendly.		☐ LCDs require mercury for their products causing harm to the environment.
• LED uses gallium arsenide phosphide.		☐ LCD uses liquid crystals and glass electrodes.

Touch Screen



A touch-sensitive LED, typically works by detecting changes in capacitance or resistance when a user touches a specific area. This change is then used to trigger a circuit that turns the LED on or off, or potentially modifies its brightness or color.

Touch-sensitive LEDs leverage the electrical properties of the human body to create a simple and intuitive user interface

Capacitive Sensing

- Many touch-sensitive LEDs utilize the principle of capacitance.
- The human body acts as a capacitor. When you touch a conductive surface, like a touch-sensitive LED, you add to its capacitance.
- The LED circuit has a built-in capacitor. When the capacitance changes due to a touch, it triggers a sensor that activates the LED.

Resistance Sensing

- Some touch-sensitive LEDs detect changes in electrical resistance when touched.
- The human body has a small electrical resistance. When you touch a conductive area, you introduce a change in the circuit's resistance.
- This change can be detected by a sensor, which in turn activates the LED

The principles of capacitive touch sensing.

